# Exploratory Data Analysis Report

# Delinquency Risk Prediction for Geldium

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## 1. Objective:

The objective of this analysis was to explore customer financial data, identify data quality issues, and detect patterns associated with delinquency risk to support predictive modelling and business decision-making.

## 2. Data Cleaning: Data Cleaning Steps Performed

- Removed duplicate customer records to improve dataset consistency.
- Identified and handled missing values in critical financial variables.
- Applied median imputation for missing credit score values.
- Used conditional imputation for missing loan account values based on account tenure.
- Reviewed dataset for inconsistencies and abnormal values.

## 3. Preparation: Missing Value Handling

Missing loan account values were handled using imputation. Records with non-zero account tenure were imputed using the median loan value, while records with zero tenure were assigned 0, assuming no active loan relationship.

## 4. Exploratory Data Analysis Findings:

- **Missed Payments Analysis:**

| Missed Payments | Delinquent Count |
|---|---|
| 0 | 14 |
| 1-2 | 23 |
| 3-4 | 21 |
| 5-6 | 22 |

**Insight:**

Customers with repeated missed payments demonstrated consistently higher delinquency levels, suggesting payment history is a strong risk indicator.

- **Debt to Income Ratio Analysis:**

| Debt to Income Ratio | Delinquent Count |
|---|---|
| 0.10-0.25 | 20 |
| 0.25-0.40 | 46 |
| 0.40+ | 14 |

**Insight:**
Customers with moderate to high debt to income ratio showed increased delinquency frequency, indicating financial obligations may impact repayment capacity.

- **Employment Status Analysis:**

| Employment Status | Delinquent Count |
|---|---|
| Unemployed | 18 |
| Self-Employed | 13 |
| Retired | 10 |
| Employed | 27 |
| EMP | 12 |

**Insight:**
Employment status appeared to influence delinquency behaviour, with employed and unemployed customers showing notable delinquency counts.

- **Age Group Analysis:**

| Age Group | Delinquent Customers |
|---|---|
| 18-32 | 17 |
| 33-46 | 16 |
| 47-60 | 29 |
| 61+ | 18 |

**Insight:**
Customers aged 47-60 showed the highest delinquency frequency, indicating elevated repayment risk within age segment.

- **Credit Utilization Analysis**

| Credit Utilization | Delinquent Count |
|---|---|
| 0.05-0.295 | 10 |
| 0.295-0.540 | 34 |
| 0.540-0.785 | 30 |
| 0.785+ | 6 |

**Insight:**

Higher credit utilization ranges were associated with elevated delinquency levels, suggested increased financial stress among customers.

## 5. Key Risk Indicators Identified:

The analysis identified several variables strongly associated with delinquency risk:

- Repeated missed payments
- High debt to income ratio
- High credit utilization
- Employment instability
- Certain age segments

## 6. Recommendations:

- Prioritize customers with repeated missed payments for early intervention.
- Monitor customers with high debt to income and credit utilization ratios.
- Improve data collection consistency for critical financial variables.
- Use customer segmentation to support targeted risk management strategies.

## 7. Conclusion:

The exploratory data analysis identified several behavioural and financial factors associated with delinquency risk. Variables such as missed payments, credit utilization, debt-to-income ratio, and employment status showed meaningful relationships with delinquent behaviour. Data cleaning and imputation improved dataset consistency and reliability for future predictive modelling. These findings can support risk-based customer segmentation and early intervention strategies for reducing payment defaults.